5.4.2*b*:

Solution: Consider $f(x) = x^2 \in W$ so $T(f(x)) = xf(x) = x(x^2) = x^3 \notin W$. Hence $T(W) \not\subseteq W$ so W is not T invariant.

5.4.3:

Solution:

(a) *Proof.* i). Clearly $\left\{\vec{0}\right\}$ is T-invariant as the only vector in $\left\{\vec{0}\right\}$ is $\vec{0}$ and $T(\vec{0}) = \vec{0}$ for any linear operator. Hence $T\left(\left\{\vec{0}\right\}\right) \subseteq \left\{\vec{0}\right\}$.

ii). V is trivially T-invariant as T has co-domain V since it is a linear operator and hence for all $v \in V$, $T(v) \in V$ by definition and thus $T(V) \subseteq V$.

□

(b) *Proof.* i.) Recall that $N(T) = \left\{v \in V \mid T(v) = \vec{0}\right\}$ and notice $\vec{0} \in N(T)$ since $T(\vec{0}) = \vec{0}$. It follows that for all $x \in N(T)$, $T(x) = \vec{0}$ and so $T(x) \in N(T)$. Hence $T(N(T)) \subseteq N(T)$.

ii.) Recall that $R(T) = \{T(v) \mid v \in V\}$ is the image of T. Hence for all $v \in R(T)$, $T(v) = T(T(x))$ for some $x \in V$. However, since T is a linear operator, $v = T(x) \in V$ so $T(v) \in R(T)$ and thus $T(R(T)) \subseteq R(T)$.

□

(c) *Proof.* Let $\lambda \in \mathbb{F}$ be an eigenvalue of T and E_λ be its corresponding eigenspace. Recall that $E_\lambda = \{x \in V \mid T(x) = \lambda x\}$. Then for all $x \in E_\lambda$, $T(x) = \lambda x$ by definition, and $\lambda x \in E_\lambda$ since E_λ is a subspace. Hence $T(E_\lambda) \subseteq E_\lambda$. □

5.4.6:

Solution:

(a) Notice that if $z = e_1 = (1, 0, 0, 0)$ then

$$\begin{aligned} T(e_1) &= (1, 0, 1, 1) \\ T^2(e_1) &= (1, -1, 2, 2) \\ T^3(e_1) &= (0, -3, 3, 3) \\ &= 0 \cdot e_1 - 3 \cdot T(e_1) + 3 \cdot T^2(e_1) \end{aligned}$$

Hence $T^3(e_1)$ is a linear combination of $e_1, T(e_1), T^2(e_1)$, and these three vectors are linearly independent (each is outside the span of the previous ones), and thus $\beta = \left\{e_1, T(e_1), T^2(e_1)\right\}$ is an ordered basis for the T-cyclic subspace generated by z.

(b) Notice that if $z = x^3$ then

$$T(z) = 6x \quad \text{and} \quad T^2(z) = \vec{0}$$

and $T^2(z) = \vec{0} = 0 \cdot z + 0 \cdot T(z)$ is a linear combination of $z, T(z)$. Since x^3 and $6x$ are not scalar multiples of each other, $\{z, T(z)\}$ is linearly independent and hence $\beta = \{z, T(z)\} = \left\{x^3, 6x\right\}$ is an ordered basis for the T-cyclic subspace generated by z.

5.4.9:

Solution:

(a) i.) Since $\beta = \{e_1, T(e_1), T^2(e_1)\}$ is a basis for W and $T^3(e_1) - 3T^2(e_1) + 3T(e_1) + 0 \cdot e_1 = 0$ (so $a_0 = 0$, $a_1 = 3$, $a_2 = -3$), it follows by Thm 5.21(b) that the characteristic polynomial is

$$f(t) = (-1)^3(3t - 3t^2 + t^3) = -t^3 + 3t^2 - 3t$$

ii.) We have

$$[T_W]_\beta = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & -3 \\ 0 & 1 & 3 \end{pmatrix} \Rightarrow \det([T_W]_\beta - tI) = \begin{vmatrix} -t & 0 & 0 \\ 1 & -t & -3 \\ 0 & 1 & 3-t \end{vmatrix} = -t(-t(3-t)+3) = -t^3+3t^2-3t$$

(b) i.) Since $\beta = \{z, T(z)\}$ is a basis for W and

$$T^2(z) = 0 \cdot T(z) + 0 \cdot z$$

the characteristic polynomial is

$$f(t) = (-1)^2 t^2 = t^2$$

ii.) We have that

$$[T_W]_\beta = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \Rightarrow \det([T_W]_\beta - tI) = \begin{vmatrix} -t & 0 \\ 1 & -t \end{vmatrix} = t^2$$

5.4.10:

Solution:

(a) Notice that $T = L_A$ for

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

hence we can find the characteristic polynomial for T:

$$\begin{aligned} f(t) = \det(A - tI) &= \begin{vmatrix} 1-t & 1 & 0 & 0 \\ 0 & 1-t & -1 & 0 \\ 1 & 0 & 1-t & 0 \\ 1 & 0 & 0 & 1-t \end{vmatrix} \\ &= (1-t)\begin{vmatrix} 1-t & 1 & 0 \\ 0 & 1-t & -1 \\ 1 & 0 & 1-t \end{vmatrix} \\ &= (1-t)\left(\begin{vmatrix} 1 & 0 \\ 1-t & -1 \end{vmatrix} + (1-t)\begin{vmatrix} 1-t & -1 \\ 0 & 1-t \end{vmatrix}\right) \\ &= (1-t)(-1 + (1-t)^3) \\ &= (1-t)^4 - (1-t) \\ &= (1-t)(-t^3 + 3t^2 - 3t) \\ &= (1-t) \cdot g(t) \end{aligned}$$

where $g(t) = -t^3 + 3t^2 - 3t$ is the characteristic polynomial for T_W. Hence, clearly $g \mid f$.

(b) Let γ be the standard ordered basis for $\mathcal{P}_3(\mathbb{R})$ and let $A = [T]_\gamma$, so that

$$A = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so T has the same characteristic polynomial as A:

$$f(t) = \det(A - tI) = \begin{vmatrix} -t & 0 & 2 & 0 \\ 0 & -t & 0 & 6 \\ 0 & 0 & -t & 0 \\ 0 & 0 & 0 & -t \end{vmatrix} = t^4 = t^2 \cdot g(t)$$

since $A - tI$ is upper triangular and $g(t) = t^2$ is the characteristic polynomial for T_W. Hence, clearly $g \mid f$.

5.4.16:

Solution:

(a) *Proof.* Let f be the characteristic polynomial of T and assume that f splits over $\mathbb{F}$. Let $n = \dim(V)$. Then there exist $\lambda_1, \ldots, \lambda_n \in \mathbb{F}$ (not necessarily distinct) such that

$$f(t) = (-1)^n (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n)$$

Then, let g be the characteristic polynomial of T_W for W some T-invariant subspace. By Thm 5.20 we know that $g \mid f$, i.e., $f(t) = g(t)h(t)$ for some polynomial h. Every irreducible factor of g is then an irreducible factor of f, and by unique factorization the irreducible factors of f are all linear. Hence g is a product of linear factors,

$$g(t) = (-1)^k (t - \lambda_{i_1})(t - \lambda_{i_2}) \cdots (t - \lambda_{i_k})$$

where $k = \dim(W)$ and each $\lambda_{i_j} \in \{\lambda_1, \ldots, \lambda_n\}$. Thus g also splits over $\mathbb{F}$. □

(b) *Proof.* Let W be a nontrivial T-invariant subspace of V and let g be the characteristic polynomial of T_W. Since $W \neq \{0\}$, $\deg(g) = \dim(W) \geq 1$, and g splits by (a), so g has at least one linear factor $t - \lambda$ and hence a root $\lambda \in \mathbb{F}$. Then λ is an eigenvalue of T_W, so there is a nonzero $w \in W$ with $T_W(w) = \lambda w$. Since $T(w) = T_W(w) = \lambda w$, w is an eigenvector of T contained in W. □

5.4.18:

Solution:

(a) *Proof.* Assume A is invertible. Equivlanently, $\det(A) \neq 0$. Recall that the characteristic polynomial is $f(t) = \det(A - tI)$ so that $f(0) = \det(A)$. However, $f(0) = a_0$ and so $a_0 = \det(A) \neq 0$.
Conversely assume $a_0 \neq 0$. As before consider $f(0) = \det(A)$ so that $\det(A) = f(0) = a_0 \neq 0$ and thus A must be invertible. □

(b) *Proof.* Assume that A is invertible, so that $a_0 \neq 0$ by (a), and let f be the characteristic polynomial

for A. Then, by the Cayley-Hamilton Theorem we know that $f(A) = O$ so

$$\begin{aligned} f(A) &= (-1)^n A^n + a_{n-1}A^{n-1} + \cdots + a_1 A + a_0 I_n = O \\ &\Rightarrow (-1)^n A^{n-1} + a_{n-1}A^{n-2} + \cdots + a_2 A + a_1 I_n + a_0 A^{-1} = O \qquad \text{(Right multiply by } A^{-1}\text{)} \\ &\Rightarrow (-1)^n A^{n-1} + a_{n-1}A^{n-2} + \cdots + a_2 A + a_1 I_n = -a_0 A^{-1} \\ &\Rightarrow -\frac{1}{a_0}\left[(-1)^n A^{n-1} + a_{n-1}A^{n-2} + \cdots + a_2 A + a_1 I_n\right] = A^{-1} \end{aligned}$$

□

(c) Notice that A is upper triangular and hence

$$\begin{aligned} f(t) = \det(A - tI) &= (1-t)(2-t)(-1-t) \\ &= (2 - 3t + t^2)(-1-t) \\ &= -t^3 + 2t^2 + t - 2 \end{aligned}$$

Hence $a_0 = -2$, $a_1 = 1$, $a_2 = 2$. Now, we have that

$$A^2 = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 6 & 6 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

so

$$A^{-1} = \frac{1}{2}\left((-1)^3 A^2 + 2A + I_3\right) = \boxed{\frac{1}{2}\begin{pmatrix} 2 & -2 & -4 \\ 0 & 1 & 3 \\ 0 & 0 & -2 \end{pmatrix}}$$

5.4.21:

Solution:

Proof. Let T be a linear operator on the two-dimensional space V and for $v \in V$ let W_v be the T-cyclic subspace generated by v. Since $W_v \subseteq V$, $\dim W_v \leq 2$ for every v. Either there exists some $v \in V$ with $\dim W_v = 2$, or $\dim W_v \neq 2$ for every $v \in V$.
Case 1 (there exists $v \in V$ with $\dim W_v = 2$): Then W_v is a 2-dimensional subspace of the 2-dimensional space V, so $W_v = V$ and V is a T-cyclic subspace of itself.
Case 2 ($\dim W_v \neq 2$ for every $v \in V$): Let $v \in V$ be nonzero. Then $\dim W_v \geq 1$ since $v \in W_v$, so $\dim W_v = 1$ and $\{v\}$ is a basis for W_v. Since $T(v) \in W_v$, we have $T(v) = c_v \cdot v$ for some $c_v \in \mathbb{F}$. Now, let $\{a, b\}$ be a basis for V. Then $a + b \neq 0$ and, by linearity of T, $T(a) + T(b) = T(a + b)$ so

$$c_a \cdot a + c_b \cdot b = c_{a+b}(a + b) \Rightarrow 0 = a(c_a - c_{a+b}) + b(c_b - c_{a+b}).$$

Since a, b are linearly independent it follows that $c_a = c_{a+b} = c_b$. Call this common scalar c. Then $T(a) = ca$ and $T(b) = cb$, so for any $u = \alpha a + \beta b \in V$,

$$T(u) = \alpha T(a) + \beta T(b) = c(\alpha a + \beta b) = cu$$

by linearity of T. Hence $T = cI$. □